INSTITUTE OF THERMOMECHANICS
CZECH ACADEMY OF SCIENCES

texMathGroup

Particle-resolved CFD-DEM with OpenHFDIB-DEM

Hands-on 4: CFD-DEM Settings

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July 02, 2026

21st OpenFOAM Workshop

Co-funded by the European Union

Ministry of Education, Youth and Sports of the Czech Republic

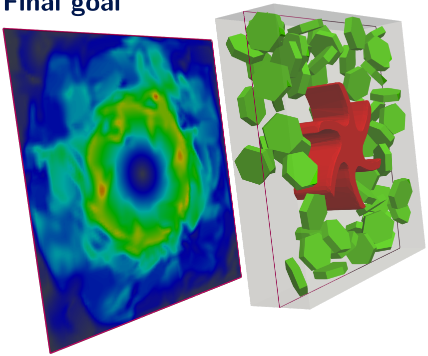
METEX
metamaterials for extremely thermally stressed components

CZ.02.01.01/00/23_020/0008501

Overal course outline

OpenHFDIB-DEM – particle-resolved CFD-DEM

Final goal



General Info | Body Addition | Movement Modes | DEM | CFD-DEM

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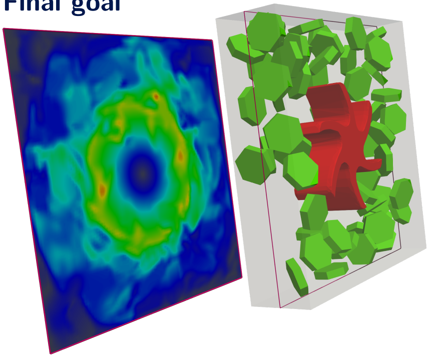

General Info → Body Addition → Movement Modes → DEM → **CFD-DEM**

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Where are we in the workflow

The parts relevant to tutorial are in green

```
case/ ─┬─ 0/ ─┬──────── U, p, ...
       │      ├──────── lambda
       │      └──────── f
       ├─ constant/ ─┬─ transportProperties
       │             ├─ turbulenceProperties
       │             ├─ polyMesh/
       │             └─ HFDIBDEMDict
       ├─ system/ ─┬─── controlDict
       │           ├─── fvSchemes
       │           └─── fvSolution
       └─────────────── Allrun
```

HFDIBDEMDict

- interpolation schemes for immersed boundary
- solid bodies definitions
  - body names
  - body types (sphere, convex, non-convex)
  - body motion (static, prescribed translation/rotation, fully coupled with fluid)
  - body addition models (addModels definition)
- DEM settings
  - DEM time step
  - solid materials
  - collision patches
  - contact treatment (virtual mesh)
- library output controls

Where to look if you are interested in the code

The parts relevant to tutorial are in green

- openHFDIB-DEM/
  - applications/
    - incompressible
    - pureDEM
    - multiphase
  - src/
    - DEM engine
    - IB coupling
    - contact models
    - geometry handling
  - tutorials/
    - ready-to-run cases
  - examples/
    - verification tests
    - validation cases
    - particle geometries
  - doc/
    - API documentation

- src/HFDIBDEM/
  - addModels
  - contactModels
  - geomModels
  - ibInterpolation
  - virtualMesh
  - openHFDIBDEM.C/H
  - immersedBody.C/H
  - ...

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Glitters and Impeller

–scan me–

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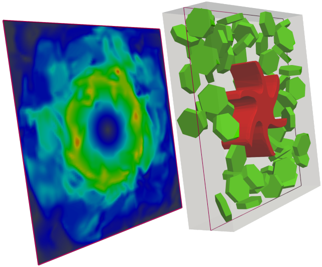
–test case presentation–



Acknowledgements

Besides the institutional support RVO:61388998, the authors acknowledge the below-listed financial support

This research was co-funded by the European Union under the project Metamaterials for thermally stressed machine components (reg. no. CZ.02.01.01/00/23_020/0008501)

METEX
metamaterials for extremely thermally stressed components

This project is co-financed from the state budget by the Czech Science Foundation (project No. 25-17815S).

FUNDED BY GAČR CZECH SCIENCE FOUNDATION

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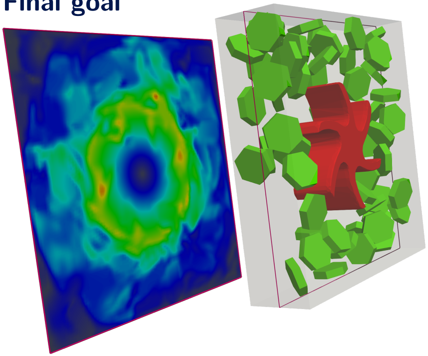

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